Math 110 Midterm 2: Barf Sheet

Polynomials

- Division: for $s \neq 0$, $p = qs + r$ with $\deg r < \deg s$ (or $r = 0$).
- $p(\lambda) = 0 \iff (x - \lambda) \mid p$. A nonzero p of degree m has at most m roots.
- FTA (Prop. 52): every nonconstant $p \in \mathcal{P}(\mathbb{C})$ has a zero; repeating, $p = c(x-\lambda_1)\cdots(x-\lambda_m)$. Over $\mathbb{R}$: linear factors times irreducible $x^2 + bx + c$ ($b^2 < 4c$). Odd-degree real p has a real root.
- p has a repeated root in $\mathbb{C} \iff p$ and p' have a common root.
- $p(T)q(T) = (pq)(T) = q(T)p(T)$: polynomials in T commute with each other and with T.
- $Tv = \lambda v \Rightarrow p(T)v = p(\lambda)v$. ▷ $T^k v = \lambda^k v$ by induction.
- $p(S^{-1}TS) = S^{-1}p(T)S$. ▷ $(S^{-1}TS)^k = S^{-1}T^kS$.

Isomorphisms (Ch. 7)

- $T\colon V \to W$ is an isomorphism if some linear S has $ST = I_V$, $TS = I_W$. The inverse is unique (Prop. 28). ▷ $S = S(TS') = (ST)S' = S'$.
- Iso $\iff$ injective and surjective (Prop. 29). ▷ Let Sw be the unique v with $Tv = w$; $T(Sw_1 + Sw_2) = w_1 + w_2$ and $T(cSw) = cw$ show S is linear.
- $V \cong W \iff \dim V = \dim W$ (finite-dim., Prop. 30). ▷ Send a basis to a basis.
- If $\dim V = \dim W < \infty$: injective $\iff$ surjective $\iff$ iso (Prop. 31). ▷ Rank–nullity: $\dim \operatorname{null} T = 0 \iff \dim \operatorname{range} T = \dim W$.
- Interpolation (Ex. 34): for distinct $x_0, \ldots, x_n$, $p \mapsto (p(x_0), \ldots, p(x_n))$, $\mathcal{P}_n(\mathbb{R}) \to \mathbb{R}^{n+1}$, is injective (a nonzero p has $\leq n$ roots), hence an iso: exactly one p has $p(x_j) = y_j$ for all j.
- Existence from injectivity (Ex. 35, LADR 3.67): $Tp = \big((x^2+5x+7)p\big)''$ maps $\mathcal{P}_m(\mathbb{R})$ to itself and is injective ($Tp = 0$ forces $\deg\big((x^2+5x+7)p\big) \leq 1$, so $p = 0$), hence surjective.
- For $S, T \in \mathcal{L}(V)$: $(ST)^{-1} = T^{-1}S^{-1}$. If ST is invertible then S is surjective and T injective; if $\dim V < \infty$ both are invertible, and so is TS (Prop. 32).
- $\dim V < \infty$, $S, T \in \mathcal{L}(V)$: $ST = I \iff TS = I$ (Prop. 33). ▷ $Tv = 0 \Rightarrow v = STv = 0$, so T is injective, hence invertible, so $S = T^{-1}$.
- Fails in infinite dim.: on $\mathcal{P}(\mathbb{R})$, $Dp = p'$, $Jp = \int_0^x p$. $DJ = I$ but $JDp = p - p(0)$. D is surjective, not injective; J injective, not surjective.
- $T \in \mathcal{L}(V)$ is an iso $\iff$ T sends every (equivalently, some) basis to a basis (HW4 P3).
- $\operatorname{rank}(BC) \leq \min(\operatorname{rank} B, \operatorname{rank} C)$ (HW4 P4). ▷ $\operatorname{range} BC \subseteq \operatorname{range} B$; $\operatorname{range} BC = B(\operatorname{range} C)$ has $\dim \leq \dim \operatorname{range} C$. Strict: $B = \left(\begin{smallmatrix}1&0\\0&0\end{smallmatrix}\right)$, $C = \left(\begin{smallmatrix}0&0\\0&1\end{smallmatrix}\right)$, $BC = 0$.
- 2×2 (HW4 P1): $\left(\begin{smallmatrix}a&b\\c&d\end{smallmatrix}\right)\left(\begin{smallmatrix}d&-b\\-c&a\end{smallmatrix}\right) = (ad-bc)I$, so $A^{-1} = \frac{1}{ad-bc}\left(\begin{smallmatrix}d&-b\\-c&a\end{smallmatrix}\right)$ when $ad \neq bc$. If $ad = bc$: a nonzero column of $\operatorname{adj} A$ is in $\operatorname{null} A$; if $\operatorname{adj} A = 0$ then $A = 0$. HW4 P2: $\left(\begin{smallmatrix}2&1\\5&3\end{smallmatrix}\right)^{-1} = \left(\begin{smallmatrix}3&-1\\-5&2\end{smallmatrix}\right)$; $\left(\begin{smallmatrix}1&i\\i&-1\end{smallmatrix}\right)$ and $\left(\begin{smallmatrix}1&2\\3&1\end{smallmatrix}\right)$ over $\mathbb{F}_5$ have determinant 0.
- V finite-dim.: an injective $S\colon U \to V$ ($U \subseteq V$) extends to an invertible $T \in \mathcal{L}(V)$ (HW4 P7). ▷ Extend a basis u_i of U and the independent list Su_i to bases of V; map the first basis to the second.
- Change of basis: if the columns of S are the new basis, the new matrix is $S^{-1}AS$. Similar matrices = one operator in two bases.

Products (Ch. 8)

- $V_1 \times \cdots \times V_m$, entrywise operations (Prop. 34); it is the direct sum of the copies $V_{(i)} = \{(0, \ldots, v_i, \ldots, 0)\}$ (Prop. 35).
- $\dim(V \times W) = \dim V + \dim W$, not the product (Prop. 36, HW4 P5). ▷ Basis $(v_i, 0), (0, w_j)$: spans since $(v, w) = (v, 0) + (0, w)$; independent by looking at each entry.
- $\Gamma\colon U_1 \times \cdots \times U_m \to U_1 + \cdots + U_m$, $(u_i) \mapsto \sum u_i$, is surjective; injective $\iff$ the sum is direct (Prop. 37). ▷ $\operatorname{null}\Gamma = \{0\}$ is the definition of a direct sum.
- $m = 2$, U_1, U_2 finite-dim.: $\operatorname{null}\Gamma = \{(u, -u) : u \in U_1 \cap U_2\} \cong U_1 \cap U_2$, so rank–nullity gives $\dim(U_1 + U_2) = \dim U_1 + \dim U_2 - \dim(U_1 \cap U_2)$. Sum direct $\iff \dim(\sum U_i) = \sum \dim U_i$.

Affine subsets and quotient spaces (Ch. 8)

- $v + U = \{v + u : u \in U\}$. An affine subset ($A = v + U$, Def. 22; parallel to U, Def. 23) is parallel to exactly one subspace (Prop. 38). ▷ $U = \{a - a' : a, a' \in A\}$.
- TFAE: $v - w \in U$; $v + U = w + U$; $(v + U) \cap (w + U) \neq \varnothing$ (Prop. 39). Two translates are equal or disjoint.
- $v + U$ is a subspace $\iff v \in U$, and then $v + U = U$ (HW4 P6).
- $V/U = \{v+U\}$; $(v+U)+(w+U) = (v+w)+U$, $\lambda(v+U) = \lambda v + U$. Well-defined: if $v - v', w - w' \in U$, then $(v+w) - (v'+w') \in U$ and $\lambda v - \lambda v' \in U$. Zero is $0 + U = U$.
- $\pi\colon V \to V/U$, $v \mapsto v + U$: linear, surjective, $\operatorname{null}\pi = U$, so $\dim V/U = \dim V - \dim U$ if V is finite-dim. (Prop. 40).
- Basis of V/U: extend a basis $u_1, \ldots, u_k$ of U by $w_1, \ldots, w_r$ to a basis of V; then $w_1 + U, \ldots, w_r + U$ is a basis. In general $v_1 + U, \ldots, v_r + U$ is a basis $\iff$ u's together with v's form a basis of V.
- $V = U \oplus W \Rightarrow W \cong V/U$ via $w \mapsto w + U$.
- If $Tx_0 = b$, the solutions of $Tx = b$ are $x_0 + \operatorname{null} T$ (affine); none if $b \notin \operatorname{range} T$. T is constant on each coset of $\operatorname{null} T$.

Isomorphism theorems and induced maps (Ch. 9)

- **First** (Prop. 41): $\tilde{T}\colon V/\operatorname{null} T \to \operatorname{range} T$, $v + \operatorname{null} T \mapsto Tv$ is an isomorphism and $T = \iota \circ \tilde{T} \circ \pi$. ▷ Well-defined: $v - w \in \operatorname{null} T \Rightarrow Tv = Tw$. Injective: $Tv = 0 \Rightarrow v + \operatorname{null} T = 0$. Surjective: clear.
- **Induced maps** (Prop. 42): if $T(U_1) \subseteq U_2$, then $v + U_1 \mapsto Tv + U_2$ is a well-defined linear map $V/U_1 \to W/U_2$. ▷ $v - w \in U_1 \Rightarrow Tv - Tw \in U_2$.
- **Universal property** (HW5 P2): if $U \subseteq \operatorname{null} T$, there is a unique $\tilde{T}\colon V/U \to W$ with $\tilde{T}(v + U) = Tv$, and $\operatorname{null}\tilde{T} = (\operatorname{null} T)/U$, $\operatorname{range}\tilde{T} = \operatorname{range} T$. Uniqueness: π is surjective.
- **Second** (Prop. 43): $U/(U \cap W) \cong (U + W)/W$, $u + U \cap W \mapsto u + W$. ▷ $U \to (U + W)/W$, $u \mapsto u + W$ is surjective ($u + w + W = u + W$) with null space $U \cap W$; apply the first.
- **Third** (Prop. 44): $W \subseteq U \subseteq V \Rightarrow (V/W)/(U/W) \cong V/U$. ▷ $V/W \to V/U$, $v + W \mapsto v + U$ is induced by I (as $W \subseteq U$), surjective, null space U/W; apply the first.
- To identify V/U: find a surjective $T\colon V \to X$ with $\operatorname{null} T = U$; then $V/U \cong X$, $v + U \mapsto Tv$. Example (PM2 Q1): $\mathcal{P}_3(\mathbb{R})/\{p : p(0) = p(1) = 0\} \cong \mathbb{R}^2$ via $p \mapsto (p(0), p(1))$.
- $T(U) \cong U/(U \cap \operatorname{null} T)$ (first theorem for $T|_U$).
- $D\colon \mathcal{P}(\mathbb{R}) \to \mathcal{P}(\mathbb{R})$: $\mathcal{P}(\mathbb{R})/\mathbb{R} \cong \mathcal{P}(\mathbb{R})$, $p + \mathbb{R} \mapsto p'$ (Ex. 43).

Dual space and dual basis (Ch. 10)

- $V' = \mathcal{L}(V, \mathbb{F})$ (Def. 26); $\dim V' = \dim V$ if V is finite-dim. Hammond's exam writes V^*, T^* for V', T'.
- A functional is 0 or surjective (HW5 P3). If $\varphi \neq 0$: $\dim \operatorname{null} \varphi = \dim V - 1$ (V finite-dim.) and $V = \operatorname{null} \varphi \oplus \operatorname{span}(v)$ for any v with $\varphi(v) \neq 0$.
- $\operatorname{null} \varphi = \operatorname{null} \psi \Rightarrow \psi = c\varphi$. ▷ If $\varphi = 0$, then $\psi = 0$. Else pick $\varphi(v) = 1$, $c = \psi(v)$; $\psi - c\varphi$ kills $\operatorname{null} \varphi$ and v, which span V.
- Dual basis: $\varphi_j(v_k) = \delta_{jk}$ (Def. 27). It is a basis of V' (Prop. 45). ▷ Apply $\sum \lambda_j \varphi_j = 0$ to v_k: $\lambda_k = 0$. Then count dimensions.
- Coordinates (HW5 P4): $v = \sum_j \varphi_j(v)\, v_j$ and $\psi = \sum_j \psi(v_j)\, \varphi_j$. ▷ Both sides agree on each v_k.
- In $\mathbb{F}^n$: if the v_j are the columns of B, the dual basis is the rows of B^{-1}. ▷ $B^{-1}B = I$ says row j applied to column k is δ_{jk}.
- Double dual $j_V(v)(\varphi) = \varphi(v)$: always injective, iso in finite dimension (Prop. 46). ▷ $v \neq 0$: extend v to a basis, take φ with $\varphi(v) = 1$; then $j_V(v)(\varphi) = 1$. Then $\dim V'' = \dim V$. Natural: $T'' \circ j_V = j_W \circ T$ (Prop. 48).

Dual maps (Ch. 10)

- $T' \colon W' \to V'$, $T'(\psi) = \psi \circ T$ (Def. 28). $T'\psi$ is linear (composition), and T' is linear: $(\psi_1 + \psi_2) \circ T = \psi_1 \circ T + \psi_2 \circ T$, $(c\psi) \circ T = c(\psi \circ T)$.
- $(S+T)' = S' + T'$, $(\lambda T)' = \lambda T'$, $(ST)' = T'S'$ (Prop. 47), $I' = I$, $(T^{-1})' = (T')^{-1}$. ▷ $(ST)'\psi = \psi \circ S \circ T = T'(S'\psi)$.
- Matrix: $\mathcal{M}(T') = \mathcal{M}(T)^T$ in the dual bases (Prop. 49, HW5 P5). ▷ $(T'\psi_j)(v_k) = \psi_j(Tv_k) = A_{jk}$, so $T'\psi_j = \sum_k A_{jk}\varphi_k$: column j of $\mathcal{M}(T')$ is row j of A.
- In coordinates: functional = row vector a; $T'(a) = aA$. $(AB)^T = B^T A^T$ follows from $(ST)' = T'S'$. Ex. 45: T = multiplication by x on $\mathcal{P}(\mathbb{R})$, $\varphi(p) = p(2)$: $T'\varphi = 2\varphi$.
- $\operatorname{null} T' = (\operatorname{range} T)^0$, $\operatorname{range} T' = (\operatorname{null} T)^0$ (Prop. 50). ▷ (i) $T'\varphi = 0 \iff \varphi(Tv) = 0$ for all v. (ii) $\subseteq$: $(T'\psi)(v) = \psi(Tv) = \psi(0) = 0$ on $\operatorname{null} T$. $\supseteq$: given $\varphi \in (\operatorname{null} T)^0$, $\psi(Tv) := \varphi(v)$ is well defined on $\operatorname{range} T$; extend it to W; then $T'\psi = \varphi$. Or count dimensions.
- T injective $\iff$ T' surjective; T surjective $\iff$ T' injective (Prop. 51). ▷ $\operatorname{range} T' = (\operatorname{null} T)^0 = V' \iff \operatorname{null} T = 0$; $\operatorname{null} T' = (\operatorname{range} T)^0 = 0 \iff \operatorname{range} T = W$.
- V, W finite-dim.: $\dim \operatorname{range} T' = \dim \operatorname{range} T$ (row rank = column rank); $\dim \operatorname{null} T' = \dim \operatorname{null} T + \dim W - \dim V$.
- Same eigenvalues: $T' - \lambda I = (T - \lambda I)'$ is invertible iff $T - \lambda I$ is. $E(\lambda, T') = (\operatorname{range}(T - \lambda I))^0$. $p(T') = p(T)'$, so $m_{T'} = m_T$.

Annihilators (Ch. 10)

- $U^0 = \{\varphi \in V' : \varphi|_U = 0\}$, a subspace of V' (Def. 29). $\dim U^0 = \dim V - \dim U$ (V finite-dim.). ▷ Basis $v_1, \dots, v_n$ with $v_1, \dots, v_m$ a basis of U: $\sum c_j \varphi_j \in U^0 \iff c_1 = \cdots = c_m = 0$, so $U^0 = \operatorname{span}(\varphi_{m+1}, \dots, \varphi_n)$.
- Finite dim.: $\{0\}^0 = V'$, $V^0 = \{0\}$; $U \subseteq W \iff W^0 \subseteq U^0$; $(U+W)^0 = U^0 \cap W^0$; $(U \cap W)^0 = U^0 + W^0$; $U = \{v : \varphi(v) = 0\ \forall \varphi \in U^0\}$.
- In $\mathbb{F}^n$: $a \in U^0 \iff a \cdot u_i = 0$ for a spanning list u_i of U. Solve the homogeneous system; expect $n - \dim U$ free variables. HW5 P6: $U = \operatorname{span}\big((1,1,0),(0,1,1)\big)$ gives $U^0 = \operatorname{span}\big((1,-1,1)\big)$.
- $\pi' \colon (V/U)' \to V'$ is injective with range U^0, so $(V/U)' \cong U^0$. Restriction $i' \colon V' \to U'$ is surjective with null space U^0, so $V'/U^0 \cong U'$.
- U invariant under $T \Rightarrow U^0$ invariant under T'. ▷ $(T'\varphi)(u) = \varphi(Tu) = 0$.

Invariant subspaces (Ch. 11)

- U is invariant if $T(U) \subseteq U$ (Def. 30). Always invariant: $\{0\}$, V, $\operatorname{null} T$, $\operatorname{range} T$, $E(\lambda, T)$, $\operatorname{null} p(T)$, $\operatorname{range} p(T)$. ▷ $p(T)$ commutes with T.
- If $ST = TS$: $\operatorname{null} S$, $\operatorname{range} S$, $E(\lambda, S)$ are T-invariant (and vice versa). ▷ $S(Tv) = T(Sv)$. So if $P^2 = P$ and $SP = PS$: $\operatorname{range} P$, $\operatorname{null} P$ are S-invariant and $V = \operatorname{range} P \oplus \operatorname{null} P$, so $\mathcal{M}(S)$ is block diagonal in a basis adapted to this sum (extra practice).
- Restriction $T|_U \in \mathcal{L}(U)$; quotient operator $T/U \in \mathcal{L}(V/U)$, $v + U \mapsto Tv + U$ (well defined because U is invariant). $p(T/U)(v+U) = p(T)v + U$.
- In a basis $u_1, \dots, u_m, w_1, \dots, w_r$ with the u's a basis of U: $\mathcal{M}(T) = \left(\begin{smallmatrix} A & B \\ 0 & C \end{smallmatrix}\right)$, $A = \mathcal{M}(T|_U)$, $C = \mathcal{M}(T/U)$ in the basis $w_j + U$. Two invariant complements $\Rightarrow$ block diagonal.
- Cyclic subspace $C(v, T) = \operatorname{span}(v, Tv, T^2v, \dots)$ is invariant; it equals $\operatorname{span}(v, \dots, T^{m-1}v)$ for the first m with $T^m v \in \operatorname{span}(v, \dots, T^{m-1}v)$, and $m \leq \dim V$ if V is finite-dim. (HW6 P6).
- Invariant subspaces (Ex. 49): $\operatorname{diag}(1,2)$: $\{0\}$, $\operatorname{span}(e_1)$, $\operatorname{span}(e_2)$, $\mathbb{R}^2$. Rotation: $\{0\}$, $\mathbb{R}^2$. $N = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$: $\{0\}$, $\operatorname{span}(e_1)$, $\mathbb{R}^2$.

Eigenvalues and eigenvectors (Ch. 11)

- $Tv = \lambda v$, $v \neq 0$ (Def. 31). $E(\lambda, T) = \operatorname{null}(T - \lambda I)$ (Def. 32). TFAE (finite dim.): λ eigenvalue; $T - \lambda I$ not injective; not surjective; not invertible.
- **Existence** (Prop. 53): $V \neq 0$ finite-dim. complex $\Rightarrow$ T has an eigenvector. ▷ Fix $v \neq 0$, $n = \dim V$: $v, Tv, \dots, T^n v$ dependent. Take a relation $p(T)v = 0$ of smallest degree $m \geq 1$. Factor $p = (z - \lambda)q$. Then $w = q(T)v \neq 0$ (else a smaller relation) and $(T - \lambda I)w = p(T)v = 0$. Fails over $\mathbb{R}$ (rotation) and in infinite dim. ($p \mapsto xp$ on $\mathcal{P}(\mathbb{C})$).
- **Independence** (Prop. 54): eigenvectors for distinct eigenvalues are independent. ▷ Smallest k with $v_k = \sum_{i<k} a_i v_i$. Apply T and subtract $\lambda_k \times$: $0 = \sum_{i<k} (\lambda_i - \lambda_k) a_i v_i$, so all $a_i = 0$ and $v_k = 0$. Contradiction.
- Ex. 50: $\sin x, \sin 2x, \dots, \sin Nx$ are independent: eigenvectors of $f \mapsto f''$ for the distinct eigenvalues $-1, -4, \dots, -N^2$.
- At most $\dim V$ distinct eigenvalues (Prop. 55). The sum of the eigenspaces is always direct.
- Eigenvalues of T = eigenvalues of $\mathcal{M}(T)$ in any basis (Prop. 56). $E(\lambda, S^{-1}TS) = S^{-1}E(\lambda, T)$ (HW6 P3): similar operators have the same eigenvalues and eigenspace dimensions.
- 2×2: λ eigenvalue $\iff \lambda^2 - (a+d)\lambda + (ad - bc) = 0$ (HW6 P4). ▷ $A - \lambda I$ not invertible $\iff$ its determinant is 0.
- Examples: $\left(\begin{smallmatrix} 0 & 1 \\ 1 & 1 \end{smallmatrix}\right)$ (Fibonacci, HW6 P5): $\lambda^2 - \lambda - 1$, $\lambda = \frac{1 \pm \sqrt{5}}{2}$. $\left(\begin{smallmatrix} 1 & 2 \\ 2 & 3 \end{smallmatrix}\right)$: $2 \pm \sqrt{5}$, no eigenvalues over $\mathbb{Q}$ (Ex. 56). Transpose on $\mathbb{F}^{2,2}$ (Ex. 52): $T^2 = I$, $E(1)$ = symmetric matrices (dim 3), $E(-1)$ = skew-symmetric (dim 1).
- T invertible: eigenvalues of T^{-1} are $1/\lambda$, same eigenvectors. λ eigenvalue of $T \Rightarrow p(\lambda)$ eigenvalue of $p(T)$.
- $P^2 = P$ (HW6 P1, extra practice): eigenvalues $\subseteq \{0, 1\}$; $\operatorname{range} P \cap \operatorname{null} P = 0$ ($w = Pu \Rightarrow w = P^2 u = Pw = 0$); $v = Pv + (v - Pv)$, so $V = \operatorname{range} P \oplus \operatorname{null} P$; $\operatorname{range} P = E(1, P)$; P is diagonalizable.
- D on $\mathcal{P}_n$: only eigenvalue 0, $E(0, D)$ = constants. On smooth functions every λ is an eigenvalue ($e^{\lambda x}$) (HW6 P2).
- Stochastic P (entries ≥ 0, each column sums to 1): every eigenvalue has $|\lambda| \leq 1$ (Prop. 57); 1 is an eigenvalue. ▷ $\|Px\|_1 \leq \|x\|_1$. $P^T \mathbf{1} = \mathbf{1}$ and P, P^T share eigenvalues. PageRank (Prop. 58; $0 < \alpha < 1$, v a probability vector): $I - \alpha P$ is invertible ($1/\alpha > 1$ is not an eigenvalue), so $r = \alpha P r + (1 - \alpha)v$ has a unique solu-

tion, and it is a probability vector. A probability vector π with $P\pi = \pi$ is a stationary distribution; HW6 P7: $\pi = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$. Google matrix $G = \alpha P + (1-\alpha)v\mathbf{1}^T$: $Gr = r$, and its other eigenvalues have modulus $\leq \alpha$.

Upper-triangular matrices (Ch. 12)

- TFAE (Prop. 59): $\mathcal{M}(T)$ upper-triangular in $v_1, \ldots, v_n$; $Tv_j \in \operatorname{span}(v_1, \ldots, v_j)$ for all j; each $\operatorname{span}(v_1, \ldots, v_j)$ is invariant.
- Upper-triangular $\Rightarrow$ eigenvalues = diagonal entries (Prop. 60). $\triangleright$ $T - \lambda I$ is upper-triangular with diagonal $a_{jj} - \lambda$. If $a_{kk} = \lambda$, $T - \lambda I$ maps $\operatorname{span}(v_1, .., v_k)$ into $\operatorname{span}(v_1, .., v_{k-1})$, so it is not injective. If no $a_{jj} = \lambda$, then each $v_j \in \operatorname{range}(T - \lambda I)$ by induction on j, so it is surjective.
- **Every operator on a finite-dim. complex V has an upper-triangular matrix in some basis** (Prop. 61, PM2 Q3). $\triangleright$ Induction. λ eigenvalue, $U = \operatorname{range}(T - \lambda I)$: invariant ($T$ commutes with $T - \lambda I$), $\dim U < \dim V$ ($T - \lambda I$ not injective, so not surjective). Triangular basis $u_1, .., u_m$ of U; extend by $v_1, .., v_r$: $Tv_k = (T - \lambda I)v_k + \lambda v_k \in \operatorname{span}(u\text{'s}, v_k)$.
- Over $\mathbb{R}$: a triangular basis exists $\iff$ m_T splits over $\mathbb{R}$. Rotation: none (the first basis vector would be an eigenvector). Lower-triangular: reverse the basis.

Diagonalizability (Ch. 12)

- TFAE (Prop. 62): diagonalizable; eigenbasis exists; $V = \bigoplus E(\lambda_i, T)$; $\dim V = \sum \dim E(\lambda_i, T)$. $\triangleright$ Eigenbasis $\Rightarrow$ $V = \sum E_i$, direct by Prop. 54. dim count $\Rightarrow$ the direct sum is all of V; union of bases of the E_i is an eigenbasis. $\dim E(\lambda, T)$ is the geometric multiplicity of λ.
- $\dim V$ distinct eigenvalues $\Rightarrow$ diagonalizable (Prop. 63). Converse false: I with $\dim V \geq 2$.
- Not diagonalizable: shift on $\mathbb{F}^n$, $n \geq 2$: $Ne_1 = 0$, $Ne_k = e_{k-1}$ (only eigenvalue 0, $E(0) = \operatorname{span}(e_1)$); shear $\left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$. Depends on the field: rotation is diagonalizable over $\mathbb{C}$, not over $\mathbb{R}$.
- Iteration (Prop. 64): $v = \sum a_j v_j$ in an eigenbasis $\Rightarrow$ $T^k v = \sum a_j \lambda_j^k v_j$. $A = PDP^{-1}$ (P = eigenvectors as columns) $\Rightarrow$ $A^k = PD^kP^{-1}$. Recurrences and Markov chains: the largest $|\lambda|$ dominates. Spectral gap: $\left(\begin{smallmatrix} 1-\varepsilon & \varepsilon \\ \varepsilon & 1-\varepsilon \end{smallmatrix}\right)$ has eigenvectors $(1, 1)$, $(1, -1)$, eigenvalues 1, $1 - 2\varepsilon$; the error shrinks like $(1 - 2\varepsilon)^k$.
- Restriction and quotient of a diagonalizable operator are diagonalizable (Ex. 60). $\triangleright$ $m_{T|_U}$, $m_{T/U}$ divide m_T.
- $T^n = I$ over $\mathbb{C}$ $\Rightarrow$ diagonalizable (Ex. 59): $x^n - 1$ has distinct roots. $T^2 = I$ over $\mathbb{R}$: $(x-1)(x+1)$.
- $ST = TS$ and T has $\dim V$ distinct eigenvalues $\Rightarrow$ each eigenvector of T is one of S ($E(\lambda, T)$ is a line and S-invariant).

Minimal polynomial (Ch. 13)

- (V finite-dim.) k = first index with $T^k \in \operatorname{span}(I, \ldots, T^{k-1})$; $T^k = \sum a_i T^i$ gives $m_T = x^k - \sum a_i x^i$ (Def. 36). Monic, $m_T(T) = 0$, smallest degree, unique.
- $q(T) = 0 \iff m_T \mid q$ (Prop. 65). $\triangleright$ $q = m_T r + s$, $\deg s < \deg m_T$; $s(T) = 0$; $I, \ldots, T^{d-1}$ independent, so $s = 0$.
- Zeros of m_T in $\mathbb{F}$ = eigenvalues (Prop. 66). $\triangleright$ $Tv = \lambda v$: $0 = m_T(T)v = m_T(\lambda)v$. Conversely $m_T = (x - \lambda)q$: $0 = q(T)(T - \lambda I)$; if $T - \lambda I$ were invertible, $q(T) = 0$ with $\deg q < \deg m_T$.
- $\deg m_T \leq \dim V$ (Prop. 67). $\triangleright$ Induction. $v \neq 0$; $q(T)v = 0$ with $\deg q \leq n$. If $q(T) = 0$ done. Else $U = \operatorname{null} q(T)$ is invariant, $0 \neq U \neq V$; $r = m_{T|_U}$, $p = m_{T/U}$: $p(T)v \in U$ for all v, so $r(T)p(T) = 0$ and $\deg m_T \leq \dim U + \dim V/U$.
- **Diagonalizable $\iff$ $m_T = (x - \lambda_1) \cdots (x - \lambda_m)$, distinct** $\lambda_i \in \mathbb{F}$ (Prop. 68). $\triangleright$ $\Rightarrow$: the product kills every $E(\lambda_i)$, and each λ_i is a root of m_T. $\Leftarrow$: $g = (x - \lambda_2) \cdots (x - \lambda_m) = (x - \lambda_1)q + s$, $s = g(\lambda_1) \neq 0$. $u = \frac{1}{s} g(T)v \in E(\lambda_1)$, $v - u = -\frac{1}{s}(T - \lambda_1)q(T)v \in \operatorname{range}(T - \lambda_1)$. The sum is direct by rank–nullity; m of the restriction to the range divides m_T; induct.
- T invertible $\iff$ $m_T(0) \neq 0$ (HW7 P3, PM2 Q4). If $m_T = x^d + \cdots + a_1 x + a_0$, $a_0 \neq 0$: $T^{-1} = -\frac{1}{a_0}(T^{d-1} + a_{d-1}T^{d-2} + \cdots + a_1 I)$. If $m_T = xq$: $q(T) \neq 0$, pick $w = q(T)v \neq 0$; $Tw = 0$.
- HW7 P3(c): $m_T = x^4 + x^3 - x^2 - 5x + 4 = (x-1)^2(x^2 + 3x + 4)$: $T^{-1} = -\frac{1}{4}(T^3 + T^2 - T - 5I)$; eigenvalues 1, $\frac{-3 \pm i\sqrt{7}}{2}$; not diagonalizable.
- $m_{T-\lambda I}(x) = m_T(x + \lambda)$; $m_{cT}(x) = c^d m_T(x/c)$ for $c \neq 0$, $d = \deg m_T$ (HW7 P4). $m_{S^{-1}TS} = m_T$; $m_{T'} = m_T$; for T invertible, $m_{T^{-1}}(x) = \frac{x^d}{a_0} m_T(1/x)$.
- U invariant $\Rightarrow$ $m_{T|_U} \mid m_T$ and $m_{T/U} \mid m_T$. $V = U \oplus W$ (both invariant) $\Rightarrow$ $m_T = \operatorname{lcm}(m_{T|_U}, m_{T|_W})$.
- 2×2 complex with single eigenvalue λ: $m = x - \lambda$ (iff $A = \lambda I$) or $(x - \lambda)^2$; diagonalizable iff $A = \lambda I$ (HW7 P5).
- p_v = monic of least degree with $p_v(T)v = 0$; $p_v \mid m_T$, and $m_T = \operatorname{lcm}$ of p_{v_i} over a basis.
- Lagrange (HW7 P7): $m_T = (x - \lambda_1) \cdots (x - \lambda_r)$, distinct $\lambda_j \in \mathbb{F}$; $p_j = \prod_{k \neq j} \frac{x - \lambda_k}{\lambda_j - \lambda_k}$; $p_j(T)$ is the projection onto $E(\lambda_j)$ along the other eigenspaces; $\sum_j p_j(T) = I$.
- 2×2: $A^2 - (a + d)A + (ad - bc)I = 0$, so m_A divides $x^2 - (a + d)x + (ad - bc)$.

Computing m_T

- Roots in $\mathbb{F}$ are the eigenvalues. If m_T splits over $\mathbb{F}$ (always over $\mathbb{C}$; also when some $\mathcal{M}(T)$ is upper-triangular), test $(x - \lambda_1)^{e_1} \cdots (x - \lambda_m)^{e_m}$, every $e_i \geq 1$, in increasing degree, total degree $\leq \dim V$. Over $\mathbb{R}$, m_T may also have irreducible quadratic factors.
- Diagonal matrix: $\prod$ over the distinct diagonal entries. Upper-triangular: roots are the diagonal entries; test powers.
- Cyclic vector: if $v, Tv, \ldots, T^{n-1}v$ is a basis ($n = \dim V$) and $T^n v = \sum c_i T^i v$, then $m_T = x^n - \sum c_i x^i$.
- Definition: find the first k with $T^k \in \operatorname{span}(I, \ldots, T^{k-1})$.
- Table ($V \neq 0$): $m_I = x - 1$; $m_0 = x$; shift on $\mathbb{F}^n$: x^n; D on $\mathcal{P}_n$: x^{n+1}; rotation on $\mathbb{R}^2$: $x^2 + 1$; shear: $(x-1)^2$; $\operatorname{diag}(1, 1, 2)$: $(x-1)(x-2)$; $\left(\begin{smallmatrix} 2 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{smallmatrix}\right)$: $(x-2)^2(x-3)$, $E(2) = \operatorname{span}(e_1)$, not diagonalizable (HW7 P6); $\left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$: $(x-1)(x-3)$ (Ex. 57); projection $P \neq 0, I$: $x^2 - x$.

Counterexamples to keep ready

- Rotation $R(x, y) = (-y, x)$ on $\mathbb{R}^2$: no eigenvalue, no invariant line, no triangular basis, $R^2 = -I$, $m_R = x^2 + 1$, $R^{-1} = -R$. Over $\mathbb{C}$: eigenvalues $\pm i$, eigenvectors $(1, \mp i)$.
- $N = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$: only eigenvalue 0, not diagonalizable, $N^2 = 0$, $m = x^2$. Has the same eigenvalues as 0 but is not similar to it.
- $\operatorname{diag}(1, 2)$: $e_1 + e_2$ is not an eigenvector. $T = I$: every nonzero vector is an eigenvector; repeated eigenvalue yet diagonalizable.
- $T = I$, $S = -I$, $V \neq 0$: $\operatorname{null}(S + T) = V \neq \operatorname{null} S + \operatorname{null} T$.
- Three distinct lines in $\mathbb{R}^2$: pairwise intersections 0, sum not direct.
- $V/U \cong V/W$ does not give $U = W$ (two lines in $\mathbb{R}^2$).
- Inclusion $\mathbb{F} \to \mathbb{F}^2$: injective, its dual is not.
- $\dim E(\lambda, T) \neq$ multiplicity of λ in m_T: I on $\mathbb{F}^2$, shear.

Math 110, Fall 2026 **Practice Midterm 2** **Prof. Hammond**

Preparation for Friday, October 30, 2026

NAME: _____ **SID:** _____

LAST NAME, OTHER NAMES (IN BLOCK CAPITALS)

This practice paper has four questions worth 40 points in total. It may take longer than 50 minutes; the purpose is to practise mathematical reasoning at approximately the level of the midterm, not to predict its questions.

NOTATION

For a linear map T, its null space and range are denoted by $\operatorname{null} T$ and $\operatorname{range} T$. All bases are ordered when coordinates or matrices are requested.

INSTRUCTIONS

1. We will be scanning your exam and using Gradescope. Please write only within the black borders of each page, and write your name in the space provided on each question page.
2. Please write a concise answer to each part in the space provided, including calculations or reasoning as well as the final answer.
3. You may quote without proof any theorem or example from the text, lecture notes, or worksheets, provided the result is clearly stated, unless the problem asks you to prove it.
4. Extra space for each question is provided on the back of the preceding page. Use that page only for work on the indicated question.
5. The back of the final page may be used for rough work on any question. It will not be graded.
6. Attempt the questions before consulting the solutions.

Extra space for QUESTION 1 ONLY

Name: _____

1. A quotient of a polynomial space. Let $V = \mathcal{P}_3(\mathbb{R})$ and define $E : V \to \mathbb{R}^2$ by $E(p) = (p(0), p(1))$. Put $U = \operatorname{null} E$.

a) (2) Prove that E is linear and surjective.

b) (3) Find a basis for U.

c) (3) Prove that the formula $\widetilde{E}(p + U) = E(p)$ defines a linear isomorphism from V/U to $\mathbb{R}^2$.

d) (2) Give a basis for V/U and specify the images of its elements under $\widetilde{E}$.

Extra space for QUESTION 2 ONLY

Name: _____

2. Dual maps. Let V, W, Z be finite-dimensional real vector spaces, and write $T : V \to W$ and $S : W \to Z$ for linear maps. Denote their dual maps by $T^* : W^* \to V^*$ and $S^* : Z^* \to W^*$.

a) (2) Give the definition of T^* and prove that it is linear.

b) (3) Prove that $(S \circ T)^* = T^* \circ S^*$.

c) (3) Prove that if T is surjective, then T^* is injective.

d) (2) Suppose $V = \mathbb{R}^2$, $W = \mathbb{R}$, and $T(x, y) = 2x - y$. If $\ell \in W^*$ is defined by $\ell(t) = 3t$, compute $T^*\ell$.

Extra space for QUESTION 3 ONLY

Name: _____

3. Upper-triangularization. Let V be a nonzero finite-dimensional complex vector space, and let $T : V \to V$ be linear. You may use the theorem proved in lectures that every operator on a nonzero finite-dimensional complex vector space has an eigenvalue. Prove, by induction on $\dim V$, that some basis of V represents T by an upper-triangular matrix.

a) (2) Establish the assertion when $\dim V = 1$.

b) (3) For $\dim V > 1$, choose an eigenvalue λ of T and put $U = \text{range}(T - \lambda I)$. Prove that U is invariant under T and that $\dim U < \dim V$.

c) (2) Explain how the induction hypothesis yields a basis $u_1, \ldots, u_m$ of U satisfying $Tu_j \in \text{span}(u_1, \ldots, u_j)$ for each j.

d) (3) Extend this to a basis $u_1, \ldots, u_m, v_1, \ldots, v_r$ of V and finish the proof using $Tv_j = (T - \lambda I)v_j + \lambda v_j$.

Extra space for QUESTION 4 ONLY

Name: _____ **4. A consequence of the**

minimal polynomial. Let V be a nonzero finite-dimensional vector space over $\mathbb{R}$, let $T : V \to V$ be linear, and let m_T denote its minimal polynomial.

a) (3) Suppose $m_T(0) \neq 0$. Prove that T is invertible by expressing its inverse as a polynomial in T.

b) (3) Suppose instead that $m_T(0) = 0$. Prove that T is not invertible. You may use the defining minimality of m_T.

c) (2) Let $R : \mathbb{R}^2 \to \mathbb{R}^2$ be given by $R(x, y) = (-y, x)$. Find m_R.

d) (2) Use (a) to express R^{-1} in terms of R, and check your answer directly.

This page can be used for rough work on any question.

Math 110 — Additional Midterm Practice

Fall 2026

An operator satisfying $T^2 = T$

Let V be a finite-dimensional real vector space, and suppose that $T : V \to V$ is linear and $T^2 = T$.

(a) (3 points) Prove that $\operatorname{range} T \cap \operatorname{null} T = \{0\}$.

(b) (3 points) Prove that $V = \operatorname{range} T + \operatorname{null} T$.

(c) (4 points) Let $S : V \to V$ be a linear operator such that $ST = TS$. Prove that both $\operatorname{range} T$ and $\operatorname{null} T$ are invariant under S. Deduce that there is a basis of V with respect to which the matrix of S is block diagonal, with one block corresponding to $\operatorname{range} T$ and the other to $\operatorname{null} T$.

Exercises

Problem 1 (Dual Space Practice)**.** Consider the vector real space $V := \operatorname{span}\{1, \sin x, \sin 2x\}$ as a subspace of $C([-\pi, \pi], \mathbb{R})$, the vector space continuous real valued functions from $[-\pi, \pi]$ to $\mathbb{R}$.

(a) Show that the linear functionals $\delta_0(f) = f(0), \delta_{\pi/2}(f) = f(\pi/2)$, and $\delta_0'(f) = \frac{df}{dx}(0)$ span, and therefor form a basis of, the dual space of V.

(b) By definition, the vectors $\{1, \sin x, \sin 2x\}$ form a basis for V. Find this list's corresponding dual basis $\{\varphi_1, \varphi_2, \varphi_3\}$ written as linear combinations of $\delta_0, \delta_{\pi/2}$, and δ_0'.

(c) Consider the linear functional $\varphi(f) = \frac{1}{\pi}\int_0^\pi f(x)\,dx$ on V. Find φ's linear combination in the basis $\{\varphi_1, \varphi_2, \varphi_3\}$ and then use this to write φ as a linear combination of the dual vectors $\delta_0, \delta_{\pi/2}$, and δ_0'.

Problem 2. Let V be a finite dimensional vector space and U a subspace of V. We have maps $i : U \to V$, $i(u) = u$ and $\pi : V \to V/U$, $\pi(v) = v + U$ which are injective and surjective respectively.

(a) Calculate $\operatorname{Null} i'$, $\operatorname{Range} i'$, $\operatorname{Null} \pi'$, and $\operatorname{Range} \pi'$ where $i' : V' \to U'$ and $\pi' : (V/U)' \to V'$ are the corresponding dual maps. (By calculate I mean express as some annihilator or other subspace).

(b) Consider the quotient space $V'/\operatorname{Range} \pi'$ and construct a map $T : U' \to V'/\operatorname{Range} \pi'$ and prove that it is an isomorphism.

Problem 3. For $p \in \mathcal{P}(\mathbb{F})$ and $T \in \mathcal{L}(V)$ with eigenpair (λ, v), show $p(T)v = p(\lambda)v$.

Problem 4. Let V denote the $\mathbb{R}$-vector space space spanned by the functions $\{1, \sin x, \cos x, \sin 2x, \cos 2x\}$, and let $d\colon V \to V$ denote the differentiation operator.

1. What are the eigenvalues of d? Is d diagonalizable?
2. Define the linear map $T\colon \mathbb{C}^2 \to \mathbb{C}^2$ by $T(x, y) \coloneqq (y, -x)$. What are the eigenvalues of T? Find a basis of $\mathbb{C}^2$ such that the matrix associated to T is upper-triangular.
3. Define the linear map $T\colon \mathbb{R}^2 \to \mathbb{R}^2$ by $T(x, y) \coloneqq (y, -x)$. Show that there is no basis of $\mathbb{R}^2$ such that the matrix associated to T is upper-triangular.

Problem 5. Find the minimal polynomial, the characteristic polynomial, the eigenvalues, and the eigenspaces of the following matrix thought of as an operator $\mathbb{R}^3 \to \mathbb{R}^3$

$$\begin{pmatrix} 3 & 1 & 1 \\ 0 & 3 & -1 \\ 0 & 0 & 1 \end{pmatrix}$$

Problem 6 (Axler 5C.7)**.** Suppose $T : V \to V$ is an operator on a finite dimensional vector space V and $v \in V$.

(a) Prove that there exists a unique smallest degree polynomial p_v with leading coefficient 1 such that $p_v(T)v = 0$

(b) Prove that the minimal polynomial of T is a multiple of p_v. (This actually gives a practical way of calculating m_T in most cases).

Problem 7. For any operator $T : V \to V$ prove that the following spaces are invariant under T: (i) $\operatorname{Null} T$, (ii) $\operatorname{Range} T$. Furthermore prove that their annihilators are invariant under T'.

Problem 8 (True/False)**.**
(T/F): If v_1, v_2 are eigenvectors of T then so is $v_1 + v_2$
(T/F): If T is injective then so is its dual map T'
(T/F): Any vector space can be written as a direct sum of eigenspaces of some operator.
(T/F): The null space of $T + S$ is always $\operatorname{Null} T + \operatorname{Null} S$.
(T/F): An operator on a real vector space always has an eigenvector.
(T/F): The dimension of the eigenspace $E(\lambda, T)$ is same as the multiplicity of $(z - \lambda)$ in the minimal polynomial.
(T/F): For an upper triangular matrix, the minimal polynomial of this operator depends on the values above the diagonal.

Homework problems

HW4 P1. Let $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$.

(a) Verify by direct multiplication that $A \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} = (ad - bc)I$.

(b) Define $\det A = ad - bc$. Prove that if $\det A \neq 0$, then A is invertible and derive the usual formula for A^{-1}.

(c) Prove conversely, without using any general determinant theory, that if $\det A = 0$, then A is not invertible.

HW4 P2. Use the result of Problem 1 to decide which of the following matrices are invertible, and find the inverse whenever it exists:

$$\begin{pmatrix} 2 & 1 \\ 5 & 3 \end{pmatrix}, \qquad \begin{pmatrix} 1 & i \\ i & -1 \end{pmatrix} \text{ over } \mathbb{C}, \qquad \begin{pmatrix} 1 & 2 \\ 3 & 1 \end{pmatrix} \text{ over } \mathbb{F}_5.$$

HW4 P3. (LADR Section 3D, Exercise 3.) Let $T\colon V \to V$ be linear and suppose V is finite-dimensional. Prove that the following are equivalent:

(a) T is an isomorphism;

(b) for every basis $v_1, \dots, v_n$ of V, the list $Tv_1, \dots, Tv_n$ is a basis of V.

HW4 P4. Suppose B and C are matrices for which BC is defined.

(a) Prove $\operatorname{rank}(BC) \leq \min\{\operatorname{rank} B, \operatorname{rank} C\}$.

(b) Give an example in which the inequality is strict even though both B and C have positive rank.

HW4 P6. Let U be a subspace of V and let $v, v' \in V$.

(a) Prove that the affine set $v + U$ is a subspace of V if and only if $v \in U$; in that case $v + U = U$.

(b) Prove that $v + U = v' + U$ if and only if $v - v' \in U$.

HW4 P7. (optional) Suppose U is a subspace of a finite-dimensional vector space V, and $S\colon U \to V$ is injective. Prove that S can be extended to an invertible operator $T\colon V \to V$.

HW5 P4(a). Let $v_1, \dots, v_n$ be a basis of a finite-dimensional vector space V, and let $\varphi_1, \dots, \varphi_n$ be its dual basis. Prove that every $v \in V$ satisfies $v = \sum_{j=1}^{n} \varphi_j(v) v_j$.

HW5 P6. Let $U = \operatorname{span}\big((1,1,0),(0,1,1)\big) \subseteq \mathbb{R}^3$.

(a) Determine the annihilator $U^0 \subseteq (\mathbb{R}^3)'$, identifying a functional with its row vector.

(b) Verify in this example that $\dim U + \dim U^0 = 3$.

HW6 P2. Let $D\colon \mathcal{P}_n(\mathbb{F}) \to \mathcal{P}_n(\mathbb{F})$ be differentiation, where $\mathbb{F}$ is $\mathbb{R}$ or $\mathbb{C}$.

(a) Determine all eigenvalues and eigenspaces of D.

(b) Explain why the answer is different if one replaces $\mathcal{P}_n$ by the space of smooth functions on $\mathbb{R}$.

HW6 P4. Let $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$. Using only the 2×2 determinant formula from Homework 4, prove that $\det(tI - A) = t^2 - (a+d)t + (ad - bc)$. Then prove that a scalar λ is an eigenvalue of A if and only if it is a root of this polynomial.

HW6 P5. For each of the following matrices, find the eigenvalues and eigenspaces over the indicated field, and decide whether the matrix has a basis of eigenvectors:

$$A_1 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad A_2 = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \qquad A_3 = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}.$$

For A_1, work first over $\mathbb{R}$ and then over $\mathbb{C}$; for A_2 and A_3, work over $\mathbb{R}$.

HW6 P6. (optional) Let $T\colon V \to V$ and let $v \neq 0$. Define the cyclic subspace $C(v,T) = \operatorname{span}(v, Tv, T^2v, \dots)$.

(a) Prove that $C(v,T)$ is T-invariant.

(b) If V is finite-dimensional, prove that there is some $m \leq \dim V$ such that

$$C(v,T) = \operatorname{span}(v, Tv, \dots, T^{m-1}v).$$

HW6 P7. (optional) Consider the column-stochastic matrix $P = \begin{pmatrix} 1/2 & 1/2 & 0 \\ 1/2 & 0 & 1/2 \\ 0 & 1/2 & 1/2 \end{pmatrix}$. Find a probability vector p satisfying $Pp = p$, and explain its interpretation for the corresponding three-state Markov chain.

HW7 P1. Return to the matrices from Homework 6, $A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$.

(a) Find the minimal polynomial of A over $\mathbb{R}$ and over $\mathbb{C}$.

(b) Find the minimal polynomial of B over $\mathbb{R}$.

(c) Use the minimal-polynomial criterion for diagonalizability to recover the conclusions of Homework 6 about whether these operators are diagonalizable over the indicated fields.

HW7 P2. Let $D\colon \mathcal{P}_3(\mathbb{R}) \to \mathcal{P}_3(\mathbb{R})$ be differentiation.

(a) Find the minimal polynomial of D.

(b) Decide whether D is diagonalizable.

HW7 P3(c). Suppose V is a finite-dimensional complex vector space and T is an invertible operator on V with minimal polynomial $m_T(x) = x^4 + x^3 - x^2 - 5x + 4$. Express T^{-1} explicitly as a polynomial in T, find the eigenvalues of T, and decide whether T is diagonalizable.

HW7 P4(b). (optional) Let m_T be the minimal polynomial of an operator T. If $c \neq 0$ and $d = \deg m_T$, prove that the minimal polynomial of cT is $c^d m_T(x/c)$.

HW7 P6. Consider $C = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}$. Find its eigenvalues, eigenspaces, and minimal polynomial. Decide whether C is diagonalizable.

HW7 P7. (optional) Let T be an operator whose minimal polynomial splits as $m_T(x) = (x-\lambda_1)\cdots(x-\lambda_r)$ with distinct roots. For a fixed j, construct a polynomial p_j such that $p_j(\lambda_j) = 1$ and $p_j(\lambda_k) = 0$ for $k \neq j$, and explain why $p_j(T)$ is the projection onto $E(\lambda_j, T)$ along the sum of the other eigenspaces.

Practice Set 1 (40 points)

1. The second isomorphism theorem. Let U and W be subspaces of a vector space V.

a) (3) Define $T\colon U \to (U+W)/W$ by $T(u) = u + W$. Prove that T is linear and surjective.

b) (2) Prove that $\operatorname{null} T = U \cap W$.

c) (2) Deduce that $U/(U \cap W) \cong (U+W)/W$. When V is finite-dimensional, deduce $\dim(U+W) = \dim U + \dim W - \dim(U \cap W)$.

d) (3) Let $V = \mathbb{R}^{2,2}$, U the upper-triangular matrices and W the symmetric matrices. Find $U \cap W$ and $U + W$. Give a basis of $U/(U \cap W)$ and the image of that basis under the isomorphism in (c).

2. Dual bases and null spaces of dual maps. Let V be finite-dimensional with basis $v_1, \dots, v_n$ and dual basis $\varphi_1, \dots, \varphi_n$.

a) (3) Prove that $\varphi_1, \dots, \varphi_n$ is a basis of V'.

b) (2) Prove that every $\psi \in V'$ satisfies $\psi = \psi(v_1)\varphi_1 + \cdots + \psi(v_n)\varphi_n$.

c) (3) Let $T\colon V \to W$ be linear, W finite-dimensional. Prove that $\operatorname{null} T' = (\operatorname{range} T)^0$, and deduce $\dim \operatorname{null} T' = \dim W - \dim \operatorname{range} T$.

d) (2) In $\mathbb{R}^2$ let $v_1 = (1,1)$, $v_2 = (1,2)$. Find the dual basis φ_1, φ_2 and write $\psi(x,y) = 3x - y$ as a combination of φ_1, φ_2.

3. Eigenvectors and diagonalizability. Let V be finite-dimensional and $T \in \mathcal{L}(V)$.

a) (4) Let $v_1, \dots, v_m$ be eigenvectors of T with distinct eigenvalues $\lambda_1, \dots, \lambda_m$. Prove that $v_1, \dots, v_m$ is linearly independent.

b) (2) Let $\lambda_1, \dots, \lambda_m$ be the distinct eigenvalues of T. Prove that the sum $E(\lambda_1, T) + \cdots + E(\lambda_m, T)$ is direct.

c) (2) Prove that T is diagonalizable if and only if $\dim V = \dim E(\lambda_1, T) + \cdots + \dim E(\lambda_m, T)$.

d) (2) Let $A = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 2 & 1 \\ 0 & 0 & 1 \end{pmatrix}$ act on $\mathbb{R}^3$. Find its eigenspaces and decide whether A is diagonalizable. If it is, give an eigenbasis.

4. Diagonalizability and the minimal polynomial. Let V be finite-dimensional and $T \in \mathcal{L}(V)$ with minimal polynomial m_T. You may use that the zeros of m_T are the eigenvalues of T.

a) (3) Suppose T is diagonalizable with distinct eigenvalues $\lambda_1, \dots, \lambda_m$. Prove that $m_T = (x - \lambda_1)\cdots(x - \lambda_m)$.

b) (3) Conversely, suppose $m_T = (x - \lambda_1)\cdots(x - \lambda_m)$ with distinct $\lambda_i \in \mathbb{F}$ and $m \geq 2$. Let $g = (x - \lambda_2)\cdots(x - \lambda_m)$ and write $g = (x - \lambda_1)q + s$ with s a constant. Prove that $s \neq 0$ and that $V = \operatorname{null}(T - \lambda_1 I) + \operatorname{range}(T - \lambda_1 I)$. (Hint: for $v \in V$ consider $u = \frac{1}{s} g(T)v$.)

c) (2) Prove that the sum in (b) is direct, that $U = \operatorname{range}(T - \lambda_1 I)$ is invariant under T, and that $m_{T|_U}$ divides m_T. Explain how induction on $\dim V$ completes the proof that T is diagonalizable.

d) (2) Let $T\colon \mathcal{P}_2(\mathbb{R}) \to \mathcal{P}_2(\mathbb{R})$, $T(p) = p' + 2p$. Find m_T and decide whether T is diagonalizable.

Practice Set 2 (40 points)

1. Isomorphisms and products.

a) (3) Let V be finite-dimensional and $S, T \in \mathcal{L}(V)$ with $ST = I$. Prove that $TS = I$.

b) (2) On $V = \mathcal{P}(\mathbb{R})$ let $D(p) = p'$ and $(Jp)(x) = \int_0^x p(t)\,dt$. Show that $DJ = I$ but $JD \neq I$.

c) (3) Let V, W be finite-dimensional with bases $v_1, \dots, v_m$ and $w_1, \dots, w_n$. Prove that the list

$$(v_1, 0), \dots, (v_m, 0), (0, w_1), \dots, (0, w_n)$$

is a basis of $V \times W$.

d) (2) Let U_1, U_2 be subspaces of a finite-dimensional V and $\Gamma\colon U_1 \times U_2 \to V$, $\Gamma(u_1, u_2) = u_1 + u_2$. Prove that $\operatorname{null}\Gamma = \{(u, -u) : u \in U_1 \cap U_2\}$, and deduce that $U_1 + U_2$ is a direct sum if and only if $\dim(U_1 + U_2) = \dim U_1 + \dim U_2$.

2. Annihilators and the double dual. Let V, W be finite-dimensional.

a) (3) Let U be a subspace of V. Prove that $\dim U^0 = \dim V - \dim U$.

b) (3) Let $T\colon V \to W$ be linear. Prove that $\operatorname{range} T' \subseteq (\operatorname{null} T)^0$, and use dimensions to prove equality. You may use $\operatorname{null} T' = (\operatorname{range} T)^0$.

c) (2) Define $j_V\colon V \to V''$ by $j_V(v)(\varphi) = \varphi(v)$. Prove that j_V is injective, and deduce that it is an isomorphism.

d) (2) Let $U = \operatorname{span}\big((1, 0, -1, 0), (0, 1, 0, -1)\big) \subseteq \mathbb{R}^4$. Identifying functionals with row vectors, find a basis of U^0.

3. Eigenvectors over $\mathbb{C}$ and upper-triangular matrices.

a) (2) Let $V \neq \{0\}$ be a complex vector space of dimension n, $T \in \mathcal{L}(V)$ and $v \neq 0$. Explain why there is a relation $a_0 v + a_1 Tv + \cdots + a_m T^m v = 0$ with $a_m \neq 0$, and why the smallest possible m in such a relation is at least 1.

b) (2) Take a relation with m smallest, let $p(z) = a_0 + a_1 z + \cdots + a_m z^m$, and factor $p(z) = (z - \lambda)q(z)$. Prove that $w = q(T)v$ is an eigenvector of T.

c) (3) Let $T \in \mathcal{L}(V)$ have an upper-triangular matrix in the basis $v_1, \dots, v_n$, with diagonal entries $\lambda_1, \dots, \lambda_n$. Prove that each λ_k is an eigenvalue of T.

d) (3) Let $T(x, y) = (x - y,\ x + 3y)$ on $\mathbb{R}^2$. Find a basis of $\mathbb{R}^2$ in which the matrix of T is upper-triangular, write down that matrix, and decide whether T is diagonalizable.

4. Basic properties of the minimal polynomial. Let V be finite-dimensional and $T \in \mathcal{L}(V)$ with minimal polynomial m_T.

a) (3) Prove that a polynomial q satisfies $q(T) = 0$ if and only if m_T divides q.

b) (3) Prove that the zeros of m_T in $\mathbb{F}$ are exactly the eigenvalues of T.

c) (2) Prove that the minimal polynomial of $T - \lambda I$ is $m_T(x + \lambda)$.

d) (2) Let A be a 2×2 complex matrix whose only eigenvalue is λ. Prove that m_A is $x - \lambda$ or $(x - \lambda)^2$, and deduce that A is diagonalizable if and only if $A = \lambda I$.

Practice Set 3 (40 points)

1. Induced maps and the third isomorphism theorem.

a) (3) Let $T\colon V \to X$ be linear and U a subspace of V with $U \subseteq \operatorname{null} T$. Prove that there is a unique linear map $\tilde{T}\colon V/U \to X$ with $\tilde{T}(v+U) = Tv$ for all v, and that $\operatorname{null}\tilde{T} = \{v+U : v \in \operatorname{null} T\}$.

b) (3) Let $W \subseteq U \subseteq V$ be subspaces. Use (a) to prove that $(V/W)/(U/W) \cong V/U$.

c) (2) Let $D\colon \mathcal{P}_3(\mathbb{R}) \to \mathcal{P}_3(\mathbb{R})$ be differentiation and $U = \mathcal{P}_0(\mathbb{R})$ the constants. Describe the map $\tilde{D}\colon \mathcal{P}_3(\mathbb{R})/U \to \mathcal{P}_3(\mathbb{R})$ from (a), its null space and its range.

d) (2) In $V = \mathcal{P}_3(\mathbb{R})$ let $U = \{p : p(0) = 0\}$ and $W = \{p : p(0) = p'(0) = 0\}$. Check $W \subseteq U$, find $\dim V/W$ and $\dim U/W$, and give an explicit isomorphism $V/U \to \mathbb{R}$.

2. Linear functionals and dual maps. Let V and W be finite-dimensional.

a) (2) Prove that every $\varphi \in V'$ is either 0 or surjective, and deduce $\dim\operatorname{null}\varphi = \dim V - 1$ when $\varphi \neq 0$.

b) (3) Let $\varphi, \psi \in V'$ with $\operatorname{null}\varphi = \operatorname{null}\psi$. Prove that $\psi = c\varphi$ for some $c \in \mathbb{F}$.

c) (3) Let $T\colon V \to W$ be linear, $B = (v_1, \dots, v_n)$ and $C = (w_1, \dots, w_m)$ bases of V and W, and $B' = (\varphi_1, \dots, \varphi_n)$, $C' = (\psi_1, \dots, \psi_m)$ the dual bases. Prove that the matrix of T' with respect to C' and B' is the transpose of the matrix of T with respect to B and C.

d) (2) Let $D\colon \mathcal{P}_2(\mathbb{R}) \to \mathcal{P}_2(\mathbb{R})$ be differentiation and $\varphi_0, \varphi_1, \varphi_2$ the dual basis of $1, x, x^2$. Compute $D'(\varphi_0)$, $D'(\varphi_1)$ and $D'(\varphi_2)$.

3. Commuting operators and similarity. Let V be finite-dimensional and $S, T \in \mathcal{L}(V)$.

a) (2) Suppose $ST = TS$ and $\lambda \in \mathbb{F}$. Prove that $E(\lambda, T)$ is invariant under S.

b) (3) Suppose $ST = TS$, $\dim V = n$, and T has n distinct eigenvalues. Prove that every eigenvector of T is an eigenvector of S, and deduce that S is diagonalizable.

c) (3) Suppose S is invertible and $R = S^{-1}TS$. Prove that $E(\lambda, R) = S^{-1}\big(E(\lambda, T)\big)$ for every λ. Deduce that R and T have the same eigenvalues, with eigenspaces of the same dimensions.

d) (2) Let $T = \operatorname{diag}(1, 2, 3)$ on $\mathbb{R}^3$. Find all $S \in \mathcal{L}(\mathbb{R}^3)$ with $ST = TS$.

4. The degree of the minimal polynomial. Let V be finite-dimensional, $T \in \mathcal{L}(V)$, $n = \dim V \geq 1$.

a) (2) Let U be a T-invariant subspace. Prove that $m_{T|_U}$ and $m_{T/U}$ both divide m_T.

b) (3) Let $v \neq 0$. Prove that there is a nonzero polynomial q of degree at most n with $q(T)v = 0$, that $U = \operatorname{null} q(T)$ is invariant under T, and that $\deg m_T \leq n$ if $q(T) = 0$.

c) (3) Suppose $q(T) \neq 0$, and assume as induction hypothesis that the degree bound holds in dimensions less than n. Let $r = m_{T|_U}$ and $p = m_{T/U}$. Prove that $r(T)p(T) = 0$, and conclude $\deg m_T \leq n$.

d) (2) Let $A = \begin{pmatrix} 0 & 0 & 6 \\ 1 & 0 & -11 \\ 0 & 1 & 6 \end{pmatrix}$. Show that e_1, Ae_1, A^2e_1 is a basis of $\mathbb{R}^3$, find m_A, and decide whether A is diagonalizable.